Ocean Accounts: One Framework, Multiple Outputs

You will build these during the clinic.

Compile once to a shared spatial framework. Report to many policy instruments.

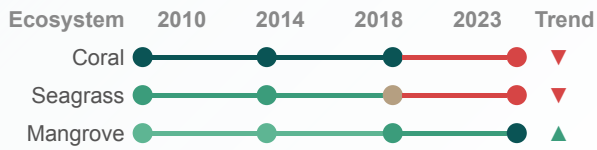
1 Extent Account (ha)

Ecosystem	Opening	Closing	Change
Coral reef	1,240	1,180	▼ −60
Seagrass	860	825	▼ −35
Mangrove	315	340	▲ +25

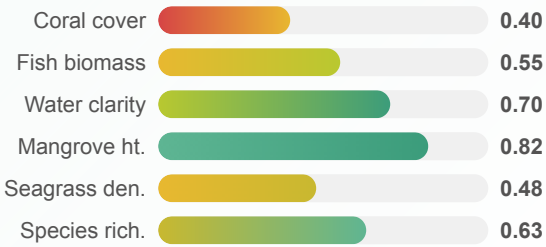
2 Ecosystem Change Matrix

	Coral	Seagrass	Mangr.	Other
Coral	1,150	10	0	30
Seagrass	5	800	0	20
Mangr.	0	0	310	5
Other	25	15	30	—

3 Extent Time Series



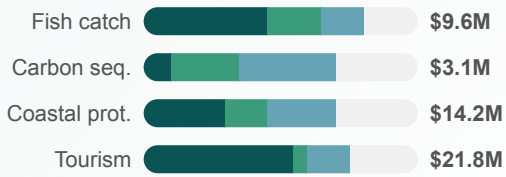
4 Condition Account (CI 0–1)



5 Services — Physical (Supply)

Service	Quantity	Unit
Fish catch	4,820	tonnes/yr
Carbon seq.	12,340	Mg CO_2/yr
Coastal prot.	18.5	km shoreline
Tourism	42,600	visitors/yr

6 Services — Monetary (USD)



All Components of One Integrated System

Compile Once, Report Many

These are not separate accounts. They are interconnected components of the same framework, sharing the same spatial data (BSUs). Economic activity depends on ecosystem services, generates waste flows, and affects social conditions.

1 Ocean Economy (OESA)

Sector	GVA (R billion)
Living resources	8.2
Minerals	5.1
Construction	12.4
Transport	28.6
Tourism	78.2
Total	**R132.5B = 2.74% GDP**

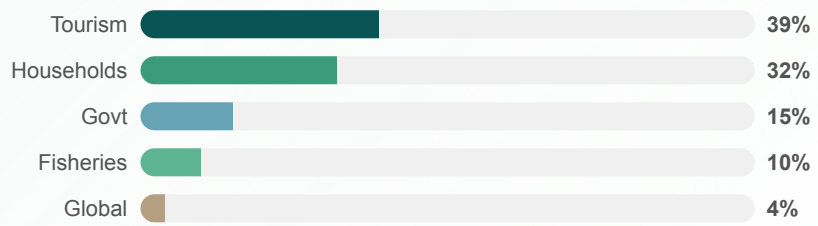
2 Waste & Emissions

N — **Water pollution**
1,200 t N/yr

S — **Solid waste**
30,000 t/yr

CO2 — **Air emissions**
800,000 t CO2/yr

3 Social Accounts

- **Human wellbeing**
- **Livelihoods**
- **Equity & Access**
- **Cultural connections**

4 Ecosystem Services USE